Hotel Booking System

Week 8 Practice - Question 4

Design: Customer has Reservations; each Room has historical reservations but no active overlapping periods. Room types set nightly rates.
Assumption: the source does not specify rates or cancellation deadline, so the example rates are Standard $80, Deluxe $130, Suite $220.
Java solution:

```
import java.time.*;import java.time.temporal.ChronoUnit;import java.util.*;
abstract class Room {final String id;Room(String i){id=i;}abstract double nightlyRate();}
class StandardRoom extends Room {StandardRoom(String i){super(i);}double nightlyRate(){return
80;}}
class DeluxeRoom extends Room {DeluxeRoom(String i){super(i);}double nightlyRate(){return
130;}}
class Suite extends Room {Suite(String i){super(i);}double nightlyRate(){return 220;}}
class Customer {final String name;Customer(String n){name=n;}}
class Reservation {final Customer customer;final Room room;final LocalDate
start,end,cancelBy;boolean active=true;
Reservation(Customer c,Room r,LocalDate s,LocalDate e,LocalDate cb){if(!e.isAfter(s))throw new
IllegalArgumentException("Checkout must be after
check-in");customer=c;room=r;start=s;end=e;cancelBy=cb;}
double price(){return ChronoUnit.DAYS.between(start,end)*room.nightlyRate();}
boolean overlaps(Reservation r){return
active&&r.active&&room.id.equals(r.room.id)&&start.isBefore(r.end)&&r.start.isBefore(end);}
boolean cancel(LocalDate today){if(!active||today.isAfter(cancelBy))return
false;active=false;return true;}}
class HotelDesk {final List<Reservation> all=new ArrayList<>();boolean available(Room
room,LocalDate s,LocalDate e){Reservation test=new Reservation(new
Customer("check"),room,s,e,s);for(Reservation r:all)if(test.overlaps(r))return false;return
true;}
Reservation reserve(Customer c,Room r,LocalDate s,LocalDate e,LocalDate
deadline){if(!available(r,s,e))throw new IllegalStateException(r.id+" is not available for
that period");Reservation x=new Reservation(c,r,s,e,deadline);all.add(x);return x;}}
public class Main {public static void main(String[] args){HotelDesk d=new HotelDesk();Room
r=new StandardRoom("Standard Room 101");LocalDate
s=LocalDate.of(2026,1,1),e=LocalDate.of(2026,1,5);
System.out.println(r.id+" is available from Jan 1 to Jan 5.");Reservation a=d.reserve(new
Customer("Customer A"),r,s,e,LocalDate.of(2025,12,30));System.out.printf("Reservation
confirmed for Customer A, Standard Room 101 (Jan 1-5). Price: $%.2f.%n",a.price());
System.out.println(r.id+" is
"+(d.available(r,LocalDate.of(2026,1,3),LocalDate.of(2026,1,7))?"available":"not available")+"
from Jan 3 to Jan 7.");if(a.cancel(LocalDate.of(2025,12,29)))System.out.println("Reservation
for Customer A, Standard Room 101 (Jan 1-5) cancelled successfully.");
Room dlx=new DeluxeRoom("Deluxe Room 201");Reservation c=d.reserve(new Customer("Customer
C"),dlx,LocalDate.of(2026,2,10),LocalDate.of(2026,2,12),LocalDate.of(2026,2,1));System.out.printf("Reservation
confirmed for Customer C, Deluxe Room 201 (Feb 10-12). Price: $%.2f.%n",c.price());}}
```

Expected example totals: Standard $320.00 for four nights, Deluxe $260.00 for two nights.
Cancellation works only on/before cancelBy.
Date overlap uses half-open [check-in, checkout) intervals, so checkout day is not charged as an overnight stay.